Conditional Manatees

Where we started and where it's going

Bayesian concepts

Simple linear regressions

DAGs and causal inference

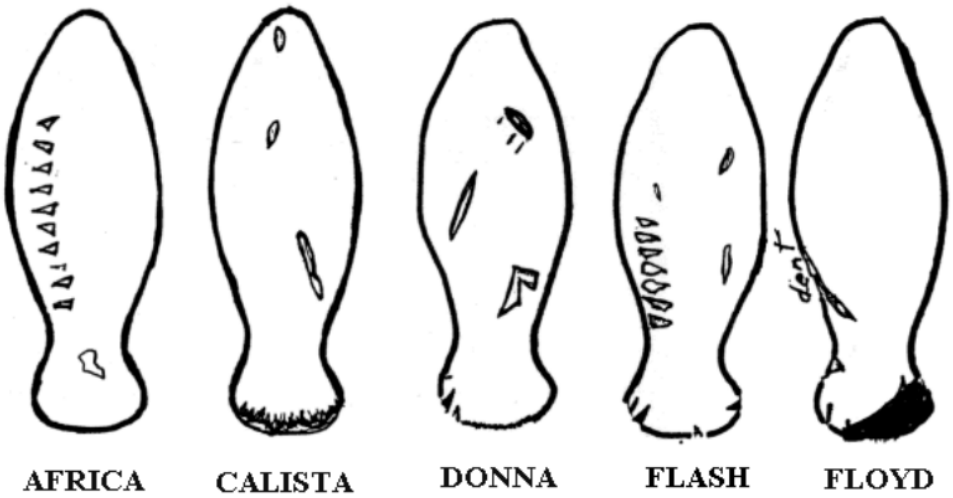
Over- and Under- fitting

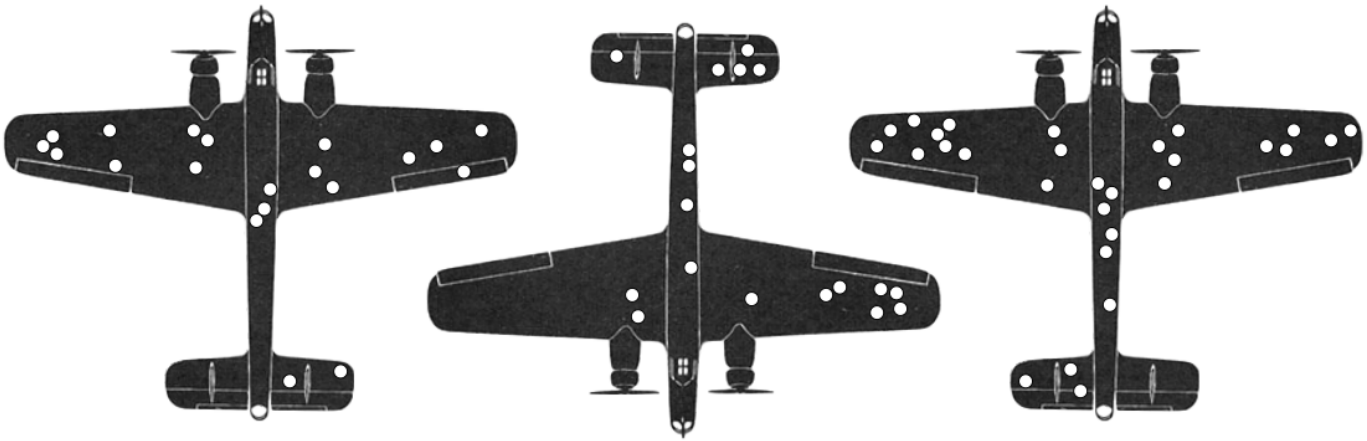
Now: Interactions

MCMC

More complex models

Conditional Manatees





Sea cows

The manatee (*Trichechus manatus*) is a slow-moving, aquatic mammal that lives in warm, shallow water.

Manatees have no natural predators, but then came people.

Majority of adult manatees bear some kind of popeller blade scar.





Letter box is not a manatee's natural habitat. Ecologists often use scar patterns to identify individuals.

Bombers

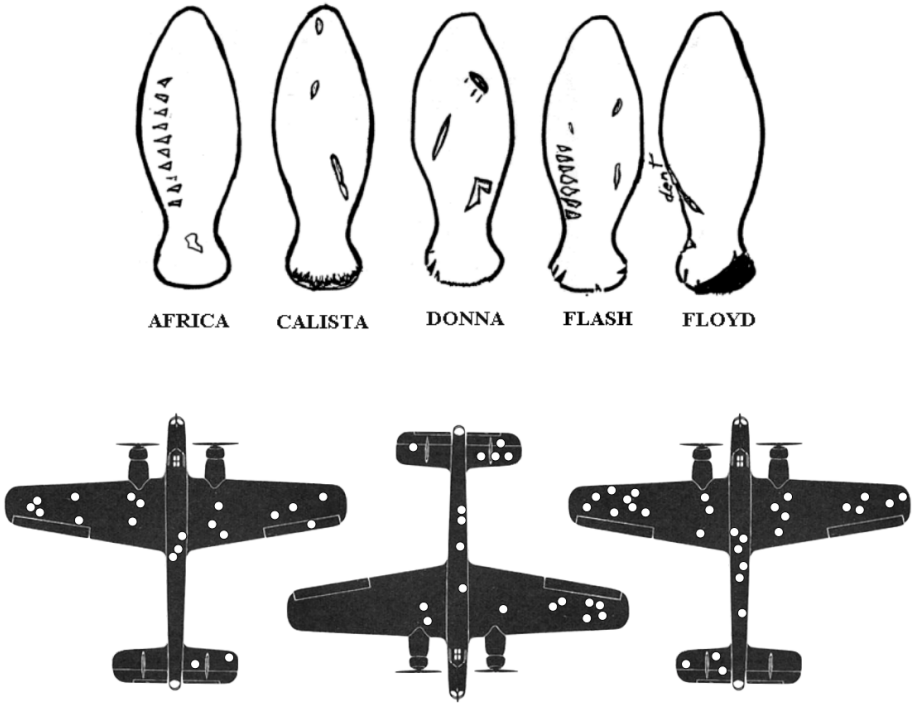
Armstrong Whitworth A.W.38 Whitley was a frontline Royal Air Force bomber.

In WWII the A.W.38 carried bombs and pamphlets into German territory.

A.W.38 has fierce natural enemies: artillery and interceptor fire.

Many planes never returned from their missions. And those that survived had the scars to prove it.

What do WWII planes and manatees have in common?



What do WWII planes and manatees have in common?

- Evidence from surviving manatees and bombers is misleading because it is conditional on surviving.
- These ones are 'lucky', manatees and bombers that perished looked different.
  -Statistics all-star, Abraham Wald

Conditioning: a key principle in statistical inference

- Data like manatees scars and bomber damages are conditional on how they got into the sample.
- Posterior distribution are conditional on the data.
- Any model-based inference is also conditional on the model.
- Every inference is conditional on something, whether we realize it or not.
- It is a strength of statistical modelling to have engines that allow probabilities to be conditional on some aspects of each case. This is the case with linear models.

Towards interactions

So far all models have assumed each predictor has an independent association with the mean of the outcome.

BUT we may want the association to be conditional

Remember the primate milk dataset:



Eulemur fulvus
0.49 *kcal/g*
55% *neocortex*



Homo sapiens
0.71 *kcal/g*
75% *neocortex*



Cebus apella
0.89 *kcal/g*
68% *neocortex*

Suppose the relationship between milk energy and brain size varies by (aka is conditional on) taxonomic group (ape, monkey, prosimian)

How do we add this to our model?

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Suppose the relationship between milk energy and brain size varies by (aka is conditional on) taxonomic group (ape, monkey, prosimian)

How do we add this to our model?
When we did multiple regressions we added another term to the model

Towards interactions

- We need a modified engine (golem).
- To model deeper conditionality -the importance of one predictor depends upon the value of at least another one- we need *interactions* (aka *moderation*).
- Interaction is a kind of conditioning allowing parameters (more specifically their posterior distribution) to depend on further aspect of the data.
- Descriptive and very powerful.
- The concepts is central beyond Gaussian outcomes and linear models of the mean.

Towards interactions

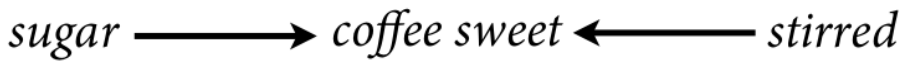
- We will allows the impact of a predictor variable to change depending upon some other variable.
- Moreover, we will estimate aspects of the distribution of those changes.
- Models with complex interactions are easy to fit to the data, thanks to their flexibility.
- Can be harder to understand.
- We will review a few simple interaction effects: how to specify them, interpret them and how to represent (plotting some predictions) them.
- A good brick to understand GLM and multilevel models.

Examples of interactions

- Interactions: influence of predictor conditional on other predictor(s),
- Examples:
  - Influence of sugar in coffee depends on stirring
  - Influence of gene on phenotype depends on environment
  - Influence of skin color on cancer depends on latitude

Interactions and DAGS

- In DAGs, interactions don't look special.

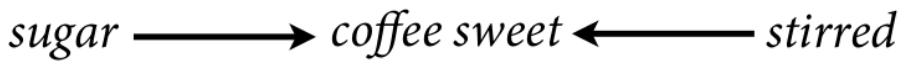


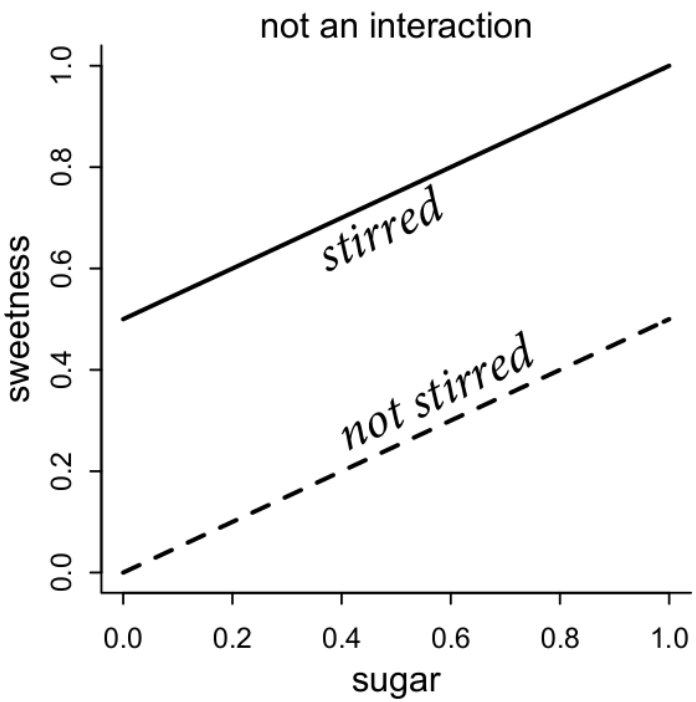
- This just means:

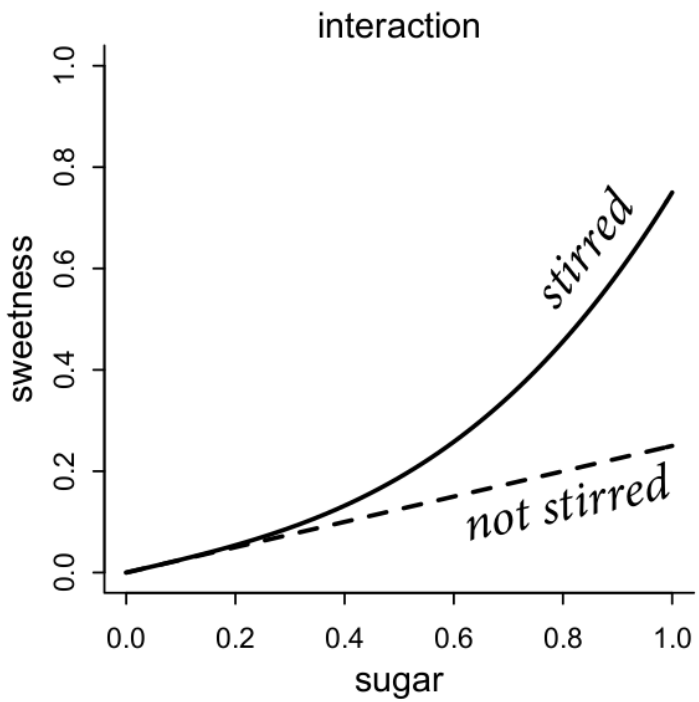
$$coffee_{sweet} = f(sugar, stirred)$$

- We have to figure out the function f.

Interactions and DAGS







Africa is special

Most culturally and genetically diverse continent

Also geographically special. . .



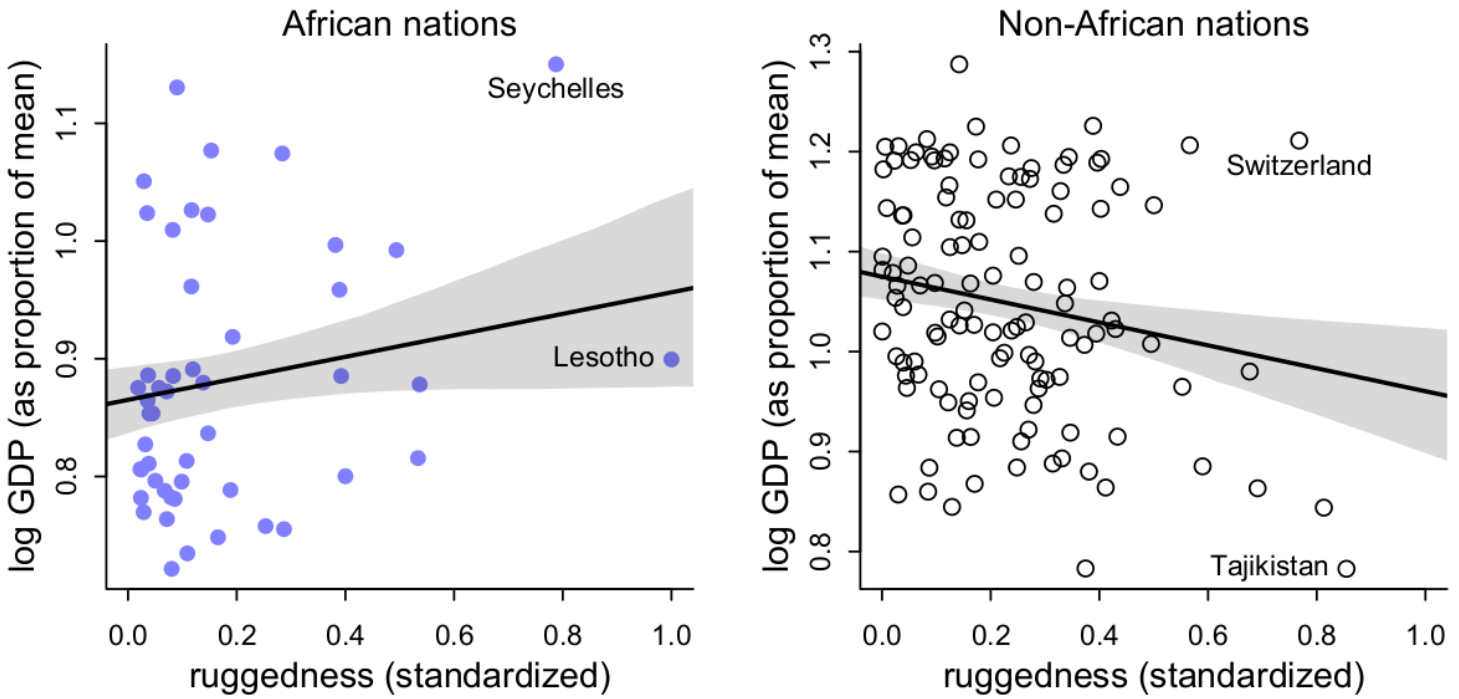
GDP vs ruggedness of different countries

```
data(rugged)
d <- rugged
```

Each row is a country and columns are economic, geographic and historical features.

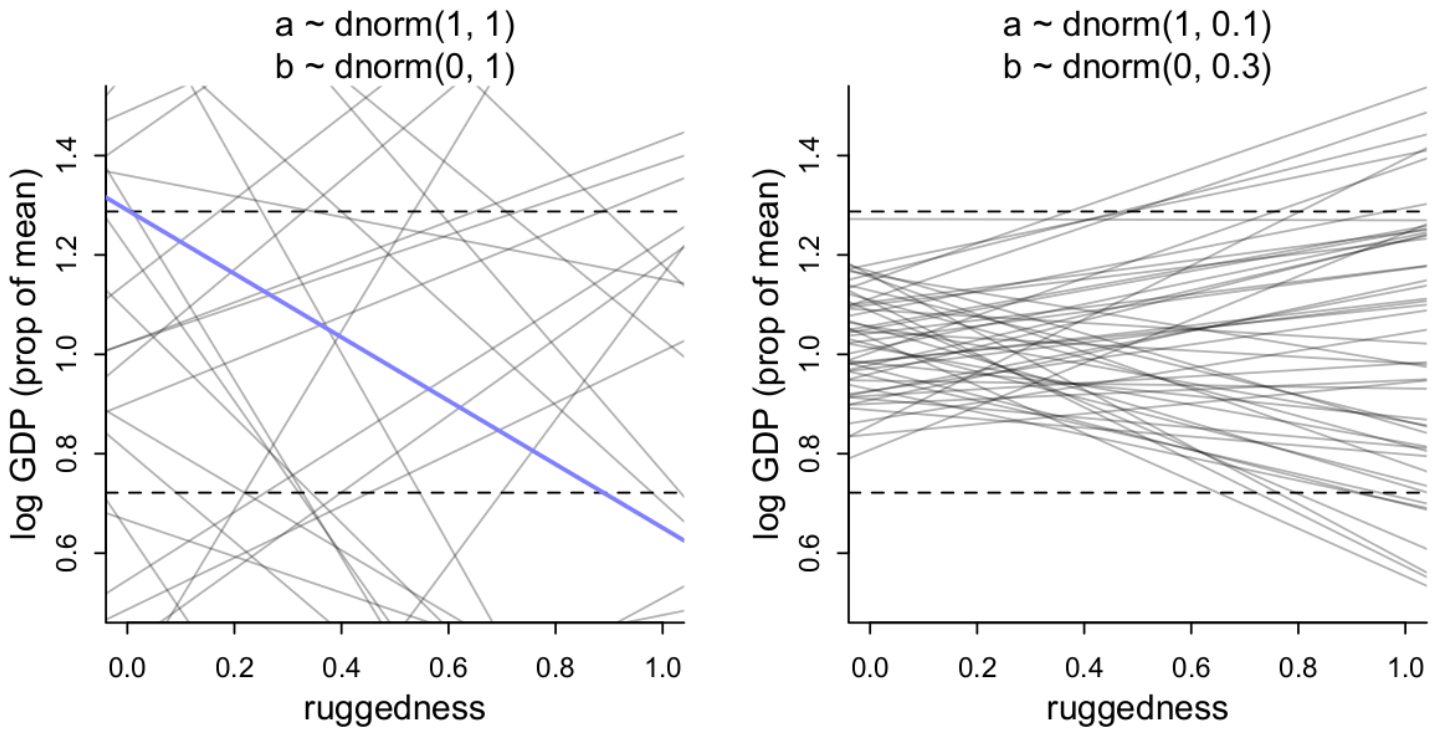
- *rgdppc_2000* outcome variable: gross domestic product per capita since 2000... we will use log (magnitude)
- *rugged* Terrain Ruggedness Index that quantifies the topographic heterogeneity of a landscape

GDP vs ruggedness of different countries



Rugged terrain -> transport is difficult -> market hampered -> reduced GDP

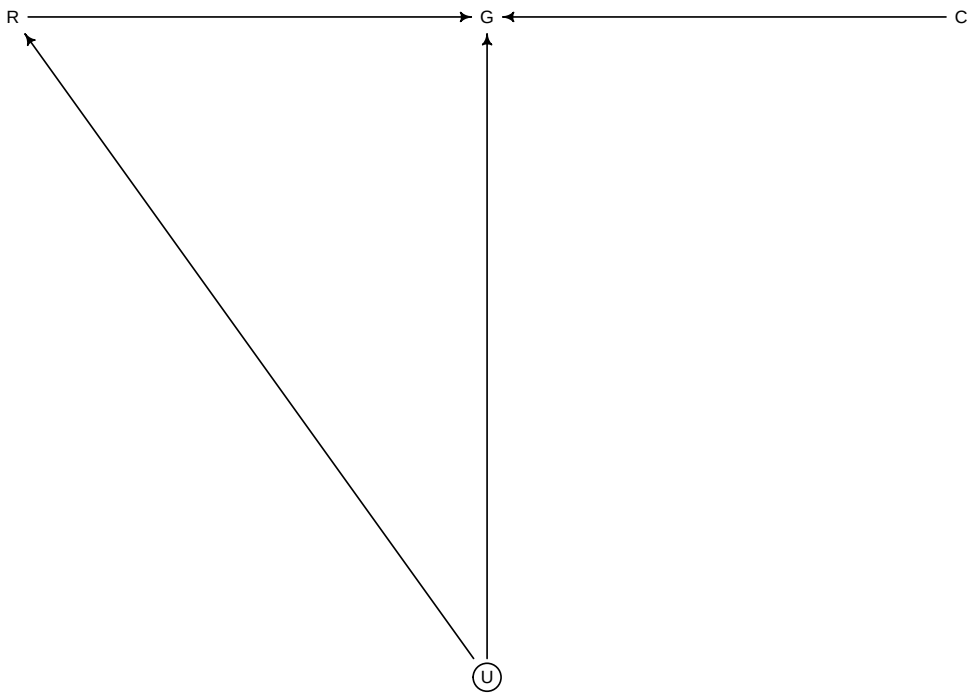
Checking and refining priors



The horizontal dashed lines show the maximum and minimum observed log GDP values so want to get the mean point $(0.215, 1.0)$.

Not crucial because simple model and lots of data. But...

GDP vs ruggedness



- Consider the DAG, focus on the implication that both R and C influence G.
- DAGs do not specify how variables combine to influence other variables
- DAG implies some function that uses R and C to generate G
  $G = f(R, C)$

Need a statistical approach to judge different propositions for $f(R, C)$

Based on what you know how could you possibly approach this problem?

We could split the data: 1 for Africa and 1 for other continents but not a good idea...

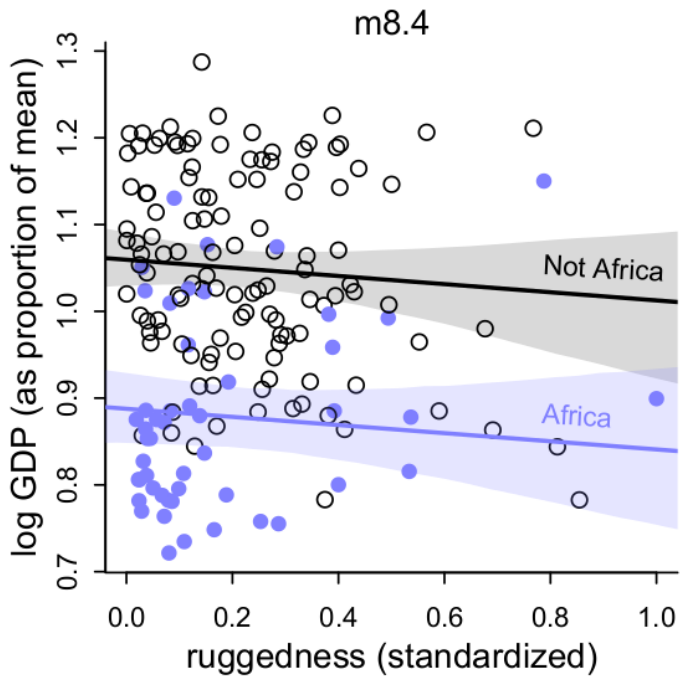
- Loosing the accuracy of the estimate ("poolable" part of the information)
- You would need to include the splitting variable into the model, otherwise, what uncertainty is associated to that split?
- If doing model comparison (via WAIC, PSIS), models need to use the same data
- Can help avoiding overfitting in each subpart by analyzing the entire data at once.

What makes Africa so special?

Something about monkeys

$$\mu_i = \alpha_{CID[i]} + \beta(r_i - \bar{r})$$

```
# make variable to index Africa (1)
# or not (2)
dd$cid <- ifelse(
            dd$cont_africa==1, 1, 2)
m8.2 <- quap(
  alist(
    log_gdp_std ~ dnorm(mu, sigma),
    mu <- a[cid] +
        b*(rugged_std -0.215),
    a[cid] ~ dnorm(1 , 0.1),
    b ~ dnorm(0 , 0.3),
    sigma ~ dexp(1)
  ), data=dd)
```



Interaction at least

- Need to allow for the effect of ruggedness to depend upon continent:

$$\mu_i = \alpha_{CID[i]} + \beta_{CID[i]}(r_i - \bar{r})$$

- It's only an index, but it means a lot :)

Interactions at least

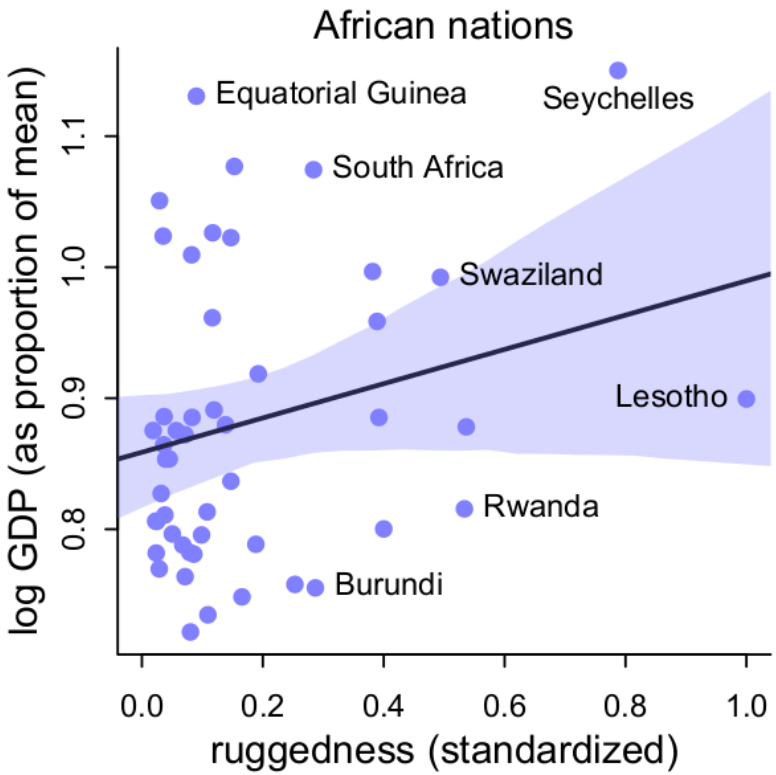
$$\mu_i = \alpha_{CID[i]} + \beta_{CID[i]}(r_i - \bar{r})$$

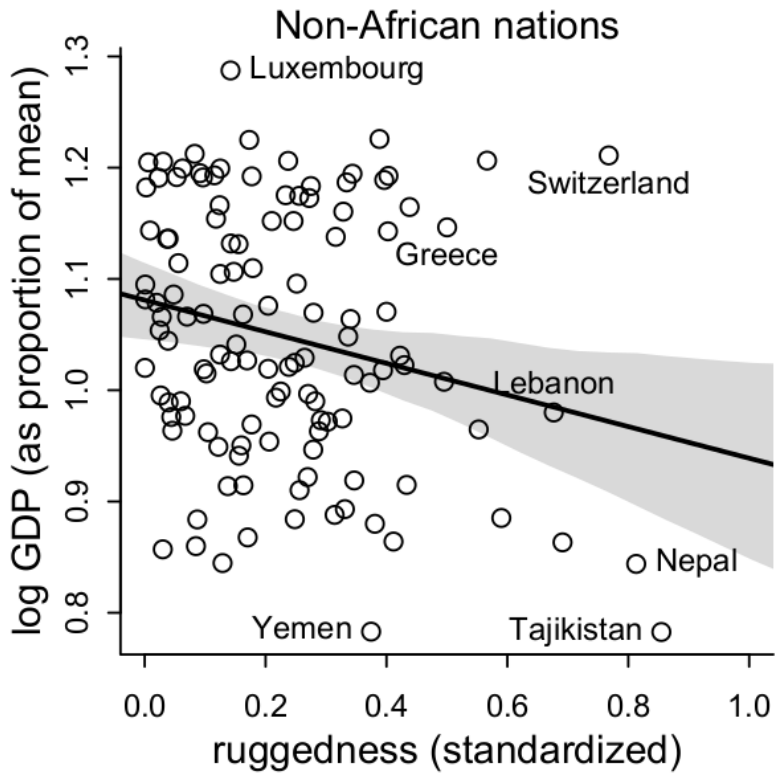
```
m8.3 <- quap(alist(
  log_gdp_std ~ dnorm(mu, sigma),
  mu <- a[cid] + b[cid]*(rugged_std - 0.215),
  a[cid] ~ dnorm(1 , 0.1),
  b[cid] ~ dnorm(0 , 0.3),
  sigma ~ dexp(1)
), data=dd)

precis(m8.3, depth = 2)
```

```
##             mean          sd        5.5%       94.5%
## a[1]   0.8865677 0.015675061  0.86151596  0.91161951
## a[2]   1.0505817 0.009936186  1.03470172  1.06646161
## b[1]   0.1325551 0.074201476  0.01396684  0.25114342
## b[2]  -0.1426909 0.054747064 -0.23018732 -0.05519456
## sigma  0.1094895 0.005934672  0.10000477  0.11897427
```

Monkey collider





Interpreting interactions

- ...is hard
  - The meaning of other parameters has changed
  - The impact of a predictor depends on multiple parameters, their covariation and the value of other predictors.

```
precis(m8.3, depth = 2)
```

```
##             mean           sd        5.5%       94.5%
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```

Interactions are symmetrical

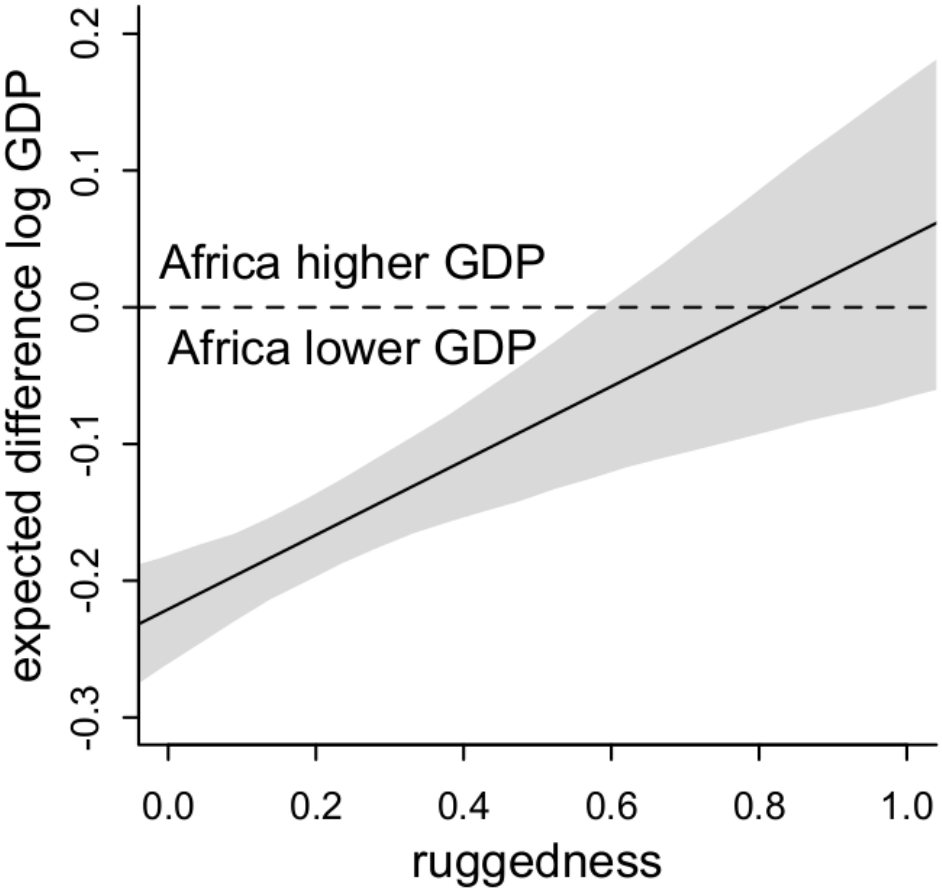
- Impact of ruggedness depends upon continent

$$\mu_i = \alpha_{CID[i]} + \beta_{CID[i]}(r_i - \bar{r})$$

- Impact of continent depends upon ruggedness

$$\mu_i = \underbrace{(2 - CID[i]) \times [\alpha_1 + \beta_1(r_i - \bar{r})]}_{CID[i]=1} + \underbrace{(2 - CID[i]) \times [\alpha_1 + \beta_1(r_i - \bar{r})]}_{CID[i]=2}$$

Interactions are symmetrical



Summary

- What do manatees and WWII planes have in common?
- Might want association between variables to be conditional
- Interactions
- Interpreting interactions

Tomorrow: More interactions